

REQUIREMENTS FOR DESIGN EXPLANATORY NOTE

2.1 General Provisions

2.1.1 The explanatory note is prepared using printing and graphic output devices of a personal computer.

When printing using a PC text editor, the **Times New Roman** font is used, font size is **13 points**, line spacing is **1,25-1.3**.

Section, subsection, paragraph, and subparagraph numbers should be bolded. Section headings are recommended to be 14-point bold, and subsection headings should be 13-point bold.

To emphasize certain elements, italic font may be used.

2.1.2 The text is placed on one side of an A4 sheet of paper, observing the margin sizes and intervals specified in Appendix L.

2.1.3 Paragraphs in the text begin with an indent of 12.5 mm (see Appendix L).

2.1.4 The explanatory note must be bound in a hard cover (a special folder for diploma projects (works)).

2.2 Headings, headings and content

2.2.1 The text of the explanatory note is divided into logically connected parts - sections, and, if necessary, into subsections, and the subsections - into points.

2.2.2 Sections should be numbered sequentially using Arabic numerals without a period at the end and written with a paragraph indent. Subsections are numbered within the section to which they belong.

2.2.3 Sometimes, within a subsection, smaller semantic units are distinguished - points, for example: characteristics of devices and functional elements of a technical system, justification of the stages of a planned experiment, characteristics of devices and instruments necessary for testing, quality indicators of a technical system in various modes of its operation, etc. In such cases, points are numbered within the subsection.

If necessary, points can be divided into sub-points, which are numbered within each point.

2.2.4 If only sections are highlighted in the explanatory note, then the points are numbered within the section.

2.2.5 Each section and subsection should have a concise and clear title. Points generally do not have titles.

Section headings are written in capital letters without a period at the end. Subsection headings are written in lowercase letters, starting with the first capital letter. Headings are not underlined. Word hyphenation is not allowed in headings. If a heading consists of two sentences, they are separated by a period.

In cases where section or subsection headings occupy several lines, the lines are aligned according to the first letter of the heading in accordance with Appendix L.

2.2.6 Each section of the explanatory note begins **on a new page**.

Between the section (subsection) heading and the text, a space is left for computer-based note-taking (see Appendix L).

Between the headings of sections and their subsections, it is permissible to place a short introductory text preceding the subsection.

2.2.7 A list of all sections and subsections, including sequential numbers and headings, is presented as a table of contents—a mandatory element of the explanatory note. The table of contents is placed immediately following the design brief and is included in the overall page numbering.

The word "Contents" is written in 14-point bold and centered on the line. A space equal to a line of whitespace is left between the word "Contents" and the content itself.

In the table of contents, each heading is connected by a dot to the page number located in the column on the right:

Content

Introduction.....	5
1 Analysis of the uncorrected management system	9
1.1 Analysis of initial data.....	9
1.2 Static and dynamic characteristics of system elements.....	12
1.3 Structural diagram of the uncorrected system.....	20
1.4 Determining the desired open-loop gain.....	21
1.5 Stability analysis.....	22
1.6 Conclusions.....	24
2 Synthesis of correction devices.....	25
Appendix A (informational) Example of filling out the title page.....	45

2.2.8 The pages of the explanatory note are numbered with Arabic numerals in the lower right corner. **The title page, summary page, and assignment sheet are included in the overall numbering, but do not include page numbers.**

2.3 Basic rules for presenting text

2.3.1 The text of the explanatory note must be clearly and logically presented and should not allow for different interpretations.

When setting out mandatory requirements, the text should use the words "must," "should," "necessary," "required," "not permitted," and "prohibited." When setting out other provisions, it is recommended to use the words "permit," "indicate," and "apply."

2.3.2 The text is presented in accordance with spelling and punctuation rules. Attention should be paid to paragraphs, enumerations, and the use of numbers, symbols, and units of measurement.

2.3.3 Small parts of the text that are separate in meaning are separated into paragraphs.

2.3.4 Explanatory notes often use enumerations, for example, in cases where it is necessary to name the composition of a technical system or a separate part of it, indicate the requirements imposed, establish the procedure for experimental research, testing, adjustment, etc.

2.3.5 If the list is simple, i.e. consists of words and phrases, then each element must be written on a new line, starting with a paragraph indent and a dash, and ending with a semicolon. For example:

The visual information reading device consists of the following units:

- distance sensor;
- photodiode matrix with dimensions of 32×32 elements;
- master oscillator and automatic sensitivity adjustment device for the photomatrix.

2.3.6 A simple listing may be written in conjunction with the text, separating words or phrases with commas. For example:

The feed drive control system uses four types of sensors: a torque sensor, a phase voltage sensor at the input of the thyristor converter, a motor shaft angular velocity sensor, and a digital load displacement sensor.

2.3.7 In a complex enumeration consisting of several sentences, each element of the enumeration is written with a capital letter. For example:

In accordance with the methodology for synthesizing digital controllers, we perform the following operations:

- We calculate the Z-transform of the transfer function of a series-connected zero-order extrapolator and the continuous part of a digital system. Using a bilinear transform, we find the corresponding characteristic in the W-transform domain.
- Using the resulting characteristic, we determine the key performance indicators of the uncorrected digital system: phase and absolute stability margins, bandwidth, resonant frequency, and resonant peak. We compare these performance indicators with the required values.
- We select a characteristic of the physically implemented controller in the W-transform domain that satisfies all control quality requirements. Finally, using the substitution $W = Z - 1/Z + 1$, we obtain the controller's transfer function in the Z-transform domain, and so on.

2.3.8 If references to enumeration elements are required in the explanatory note, they are designated using lowercase English letters with brackets. For example:

1.7 Based on the analysis of the initial data for the design of the automatic control system, we establish the following requirements for the system:

- a) the dynamic steady-state control error δy should be no more than 12 arc min at a constant speed $\Omega = 0.16$ rad/s and acceleration $\varepsilon = 0.24$ rad/s²;
- b) the moment component of the error δM should be no more than 1.2 arc min at $M_N = 104$ Nm;
- c) transition time $t_{p,p}$ no more than 0.32 s;
- d) overshoot should be no more than 24%.

When further detailing the listings, Arabic numerals with brackets are used, and the entry is made with a paragraph indentation corresponding to the level of the listing (see Appendix L).

2.3.9 When referring to an enumeration element in the text, the word “item” or “subitem” should be written without abbreviation and the bracket after the number or letter should be omitted.

For example:

In accordance with the moment error specified in paragraph 1.7 b, we determine the quality factor of the control system.

2.3.10 In the text of the explanatory note (except for formulas, tables and figures) the following should be written in words:

- the mathematical minus sign (–) before negative values of quantities;
- mathematical signs $> < =$, as well as signs №, %, □, sin, cos, etc. without numerical values, for example: “By equating the derivative of a functional to zero, we find the equation...”.

2.3.11 In text, numbers from one to nine without units of measurement should be written in words; numbers over nine should be written as digits. Fractional numbers should be given as decimals.

2.4 Basic rules for writing mathematical formulas

2.4.1 Formulas are usually placed on separate lines, centered, and separated from the text by spaces.

2.4.2 Short, similar formulas may be placed on one line, separated by semicolons. For example:

Coordinates x_1 And x_2 we determine from the known values X and Y:

$$x_1 = \Xi - \beta\Psi + \beta^2Y; \quad x_2 = \Psi - \beta Y.$$

Simple and short formulas for intermediate and auxiliary expressions can be placed directly in a line of text. Moreover, the space between lines of text can be increased.

2.4.3 If necessary, it is permissible to carry part of a mathematical expression to the next line. The operation sign at which the carryover occurs is written twice—at the end of the first line and at the beginning of the second line. When carrying a formula at the multiplication sign, the "×" sign is used instead of "·."

Hyphenations are not allowed on the division sign, as well as expressions related to the signs of the root, integral, logarithm, trigonometric functions, etc.

2.4.4 All formulas located on separate lines are numbered. A single number also denotes a group of similar formulas located on a single line.

Formulas are recommended to be numbered within the section to which they belong. The formula number should consist of the section's serial number and the formula's serial number separated by a period, for example: formula (2.7). If a section contains a single formula, it is also numbered, for example: formula (1.1).

2.4.5 The ordinal number of the formula is written in Arabic numerals in parentheses at the right edge of the line.

When transferring part of a formula from one line to another, the number is placed on the last line.

The number of a complex formula (in the form of a fraction) is written so that

the middle of the number is located at the level of the fraction line.

References in the text of the explanatory note to the ordinal number of the formula should be given in parentheses with the obligatory indication of the words “formula”, “equation”, “expression”, “equality”, “transfer function”, etc. For example:

Substituting expression (3.6) into equation (3.2), we obtain...

After the formula, a list and explanation of the symbols given that were not explained earlier should be placed.

The list begins with the word "where," which is placed on a new line without a paragraph indent; no colon is placed after the word "where." The first explanatory symbol is placed on the same line. Symbols must be separated from their explanations by a dash, with the list aligned by symbols. Each explanation ends with a semicolon. The dimension of the symbol or coefficient is indicated at the end of the explanation and separated by a comma. For example:

When accelerating the mechanism to high speed, the engine must develop a dynamic torque M_{din} , H_m , which is determined by the formula

$$M_{din} = (1.2 J_{dv} + J_{mx}) \varepsilon_{dv}, \quad (2.7)$$

where 1.2 – coefficient taking into account the reduced moment of inertia of the gearbox;

J_{dv} – moment of inertia of the engine, kg m^2 ;

J_{mx} – moment of inertia of the mechanism reduced to the engine shaft, kg m^2 ;

ε_{dv} – acceleration of the engine shaft, s^{-2} .

2.5 Basic requirements for illustrations

2.5.1 Each illustration should be clear, concise in concept, and related to the text, and placed as close as possible to the explanatory part.

It is permissible to place illustrations at the end of the explanatory note as an appendix.

2.5.2 All illustrations, regardless of their type and content, are usually called **drawings** in technical literature.

A figure should be placed after the paragraph that first references it. Multiple figures may be placed on a separate sheet. In this case, the sheet should be placed after the page that references the last figure.

An illustration placed in the text between paragraphs is centered and separated from the text and the caption by a space of one line.

2.5.3 The illustration should be positioned in such a way that it can be conveniently viewed without rotating the explanatory note or with a 90° rotation clockwise.

2.5.4 Each figure is accompanied by a caption. The caption must contain the word "Figure" without abbreviation and the sequential number of the illustration in Arabic numerals, for example: "Figure 7" for continuous numbering or "Figure 2.7" for numbering illustrations by section of the explanatory note.

An illustration must have a title, written after the number, separated by a dash and capitalized. Do not include periods after the number or title, for example:

Figure 2.1 – Structural diagram of the uncorrected control system

The signature and title are placed in alignment with the center of the drawing.

2.5.5 The text of the explanatory note must provide references to all illustrations without exception.

In references, it is recommended to use phrases such as “in accordance with Figure 2,” “Figure 5.1 shows...,” etc.

A drawing is usually completed on a single page. If a drawing doesn't fit on one page, it may be moved to other pages. In this case, the captions for the second, third, and subsequent sections of the image repeat the caption "Figure" and the illustration number, followed by the words "Sheet 2," "Sheet 3," and so on.

2.5.6 Throughout the explanatory note, uniformity must be maintained in the execution of illustrations, the design of captions, all inscriptions, dimension and extension lines, and the use of symbols.

2.6 Building tables

2.6.1 Tables are used to simplify the presentation of a text containing a fairly large volume of factual material, to give this material a more compact, convenient form for analysis and calculations, in order to increase the validity and reliability of the decisions made.

The following are usually presented in table form:

- reference information;
- function values used in graphical calculation methods;
- data from experimental studies of functional elements and devices, which are used to determine their static and dynamic characteristics; results of mathematical modeling of technical systems with automatic control, etc.

Depending on its size, it is recommended to place the table immediately after the paragraph in which it is first referenced, or on the following page. If necessary, it may be presented as an appendix to the explanatory note.

2.6.2 All tables in the text must be numbered with Arabic numerals and have a text title, with the word "table" not abbreviated. The table number and title are separated by a dash. The word "Table" begins at the left border of the table.

It is recommended that tables be numbered in accordance with the accepted system of numbering formulas and figures, for example: "Table 2" for continuous numbering or "Table 1.2" for numbering by sections of the explanatory note.

The title should be concise and accurately reflect the table's content. Header rows should not extend beyond the right and left borders of the table.

2.6.3 Tables are designed in accordance with Figure 2.1.

The table and its title are separated from the preceding and following text by a space. The title and the table itself are not separated by a space.

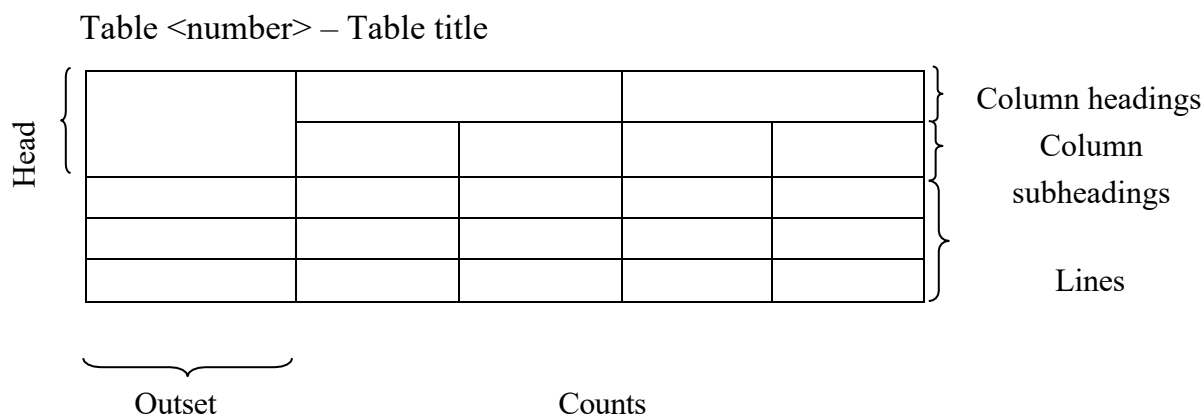


Figure 2.1 – Example of table structure (to section 2.6.3)

2.6.4 It is recommended to use lines on the left, right, and bottom of tables. If the table does not end at the end of the page, no horizontal line is used (Figure 2.2).

Table <table number> – Table title

Step	Operation code	Address of the operand	Note
0001	<i>LD</i>	0000	$PH \rightarrow P0; \Pi H \rightarrow \Sigma P1$
0002	<i>*D</i>	0001	$PH \cdot KHB \rightarrow P0; \Pi H \cdot KHB \rightarrow \Sigma P1$
...	...	...	...
0028	<i>LD</i>	1000	$PS \square R0; \square SR1$

Continuation of table <table number>

Step	Operation code	Address of the operand	Note
0029	<i>PD</i>	0010	Recording the shutter speed value in data register RD
...	...	...	...
0036	<i>WR</i>	2009	Writing a signal to a cell in memory of 2009

Figure 2.2 – Example of table formatting with continuation and repetition of the heading (to points 2.6.4, 2.6.7)

When continuing a table, the heading may be replaced by column numbers. In this case, the numbering is also placed in the first section of the table after the heading.

Subsequent parts of the table after the words “Continuation of the table...” with only its number indicated begin with a line with the numbering of the columns.

2.6.5 It is recommended that column headings be written parallel to the table rows. If necessary, perpendicular column headings are permitted (Figure 2.3).

The headings of the columns and rows of the table sidebar should be written with a capital letter, subheadings - with a lowercase letter (unless they have an independent meaning).

All headings, titles and subheadings are indicated in the nominative case singular, except in cases where the noun in the given meaning in the singular is not used in the phrase, for example: “Technical conditions”.

Words in the table should be written in full without abbreviations, with the exception of certain concepts, which may be replaced by letter designations. A period is not included at the end of the heading.

Table <table number> – Table title

State before the start operations					Operation content	Durationoperation s, with	Plannedending, with
Countd own	Robot in position	Position served					
		B	IN	G			
1	2	3	4	5	6	7	8
0.0	A				Pick up from A	0.1	0.1
0.1	A				Move to B	0.3	0.4
...	...	...	...	...	...	...	...
4.1	B	2	1		Move to B	0.3	4.4

Continuation of table <table number>

1	2	3	4	5	6	7	8
4.4	IN	2	1		Unload B	0.1	4.5
...	...	...	...	...	...	...	...
12.1	B	4	3	2	Move to G	0.6	12.7
12.7	B	4	3	2	Unload G	0.1	12.8

Figure 2.3 – An example of the design of a table and its continuation without repeating the heading (to paragraph 2.6.4)

It is prohibited to place two headings in a header cell, separated by a diagonal line, one of which refers to the sidebar, and the second combines the headings of all the columns.

The "Serial Number" column is not permitted in the table. If numbering of indicators is necessary, the serial numbers are indicated in the first column, separated by a space (Figure 2.4).

2.6.6 In a column or row of the sidebar, units of measurement of the indicator are given, separated by a comma (see Figure 2.4).

It is permissible to include in the table the column “designation of the unit of physical quantity” if most of the names in the sidebar are accompanied by dimensions.

2.6.7 If small explanations are required for most of the table rows, such explanations are presented in a separate column.

"Note" in accordance with Figure 2.2.

If it is necessary to clarify the data of individual lines or columns, then the line at the end of the table is formatted in accordance with Figure 2.4.

Table <table number> – Table title

Name of the indicator	Value of the local feedback coefficient, Ko.s			
	0.04	0.08	0.10	0.12
1 Dynamic error, arc min.	2	2.5	2.8	(3.2)
2 Transition time, s	0.10	0.105	0.11	0.12
3 Overshoot, %	(30)	25	24	22
4 Number of oscillations	(2.0)	1.5	1.5	1.5
Note: Data in brackets does not comply with technical requirements.				

Figure 2.4 – Example of formatting individual rows and columns of a table (to points 2.6.5, 2.6.7, 2.6.9)

2.6.8 A table with a small number of columns may be divided into parts and placed next to each other on one page, separated by a double line or a line of double thickness, with the table heading repeated in each part (Figure 2.5).

Table <table number> – Table title

t , min	h^* (t)	t , min	h^* (t)
0.0	0.00	3.2	0.76
0.4	0.03	3.6	0.80
0.8	0.15	4.0	0.84
1.2	0.30	4.4	0.87
1.6	0.46	4.8	0.90
2.0	0.58	5.2	0.92
2.4	0.67	6.0	0.96
2.8	0.72	6.4	0.97

Figure 2.5 – Example of a table design with a small number of columns (to points 2.6.8, 2.6.9)

2.6.9 When filling in tables, it is recommended that the number of decimal places be the same for each column of numbers in accordance with the accuracy of measuring instruments and engineering calculations.

The numerical values of one physical quantity must be arranged so that the digits of the numbers throughout the column are located one below the other (see Figure 2.5).

The values of various physical quantities are located in the middle of the cell in accordance with Figure 2.4.

If certain data is missing from a table, use a dash. Blank spaces in table columns are not permitted.

2.6.10 The explanatory note should contain brief explanations relating to the table as a whole and, if necessary, to its individual sections. These explanations should formulate the main conclusions drawn from the table's data or highlight its most characteristic or important features.

ПРИЛОЖЕНИЕ Л

(обязательное)

Размеры полей текста, расположение заголовков,
порядкового номера страницы на листе формата А4
(к пунктам 2.1.2; 2.1.3; 2.2.5; 2.2.6, 2.3.8, 2.9.1 – 2.9.3)



6 ЗАГОЛОВОК ШЕСТОГО РАЗДЕЛА

↑ Пробельная строка

6.1 Заголовок первого подраздела шестого раздела _____

6.2 Заголовок второго подраздела шестого раздела _____

Приведенный к валу двигателя момент инерции нагрузки $J_{\text{пр}}$, кг·м², вычисляем по формуле

↑ Пробельная строка

$$J_{\text{пр}} = (m_1 + m_2) \frac{v_{\frac{z}{2}}^2 \max}{v_{\frac{z}{2}}^2 \max}, \quad (6.1)$$

↑ Пробельная строка

где m_1 – _____

m_2 – _____

6.3 Техничко-экономические показатели приведены в таблице 6.1.

↑ Пробельная строка

Таблица 6.1 – Название таблицы _____

Вторая строка названия _____

1 _____ _____			
2 _____ _____			
Примечание – _____ _____			

↑ Пробельная строка

6.4 _____

Примечания

1 _____

2 _____

6.5 _____